

Recti-Q: Feature-Space Rectification for Out-of-Distribution-Robust Quantized Perception in Edge Robotics

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Abstract—Robotic perception pipelines increasingly rely on large vision backbones deployed on SWaP-constrained edge platforms, making post-training quantization (PTQ) attractive for real-time inference. However, while PTQ often preserves clean in-distribution accuracy, we show that it can substantially degrade reliability under deployment-relevant distribution shifts (e.g., sensor noise, severe weather, and novel operating environments), creating a *Quantization-Induced Robustness Gap*. Across foundational vision benchmarks (ImageNet-C and PACS), 4-bit PTQ models exhibit pronounced robustness degradation despite negligible ID accuracy loss. To address this, we propose Recti-Q, a lightweight feature-space rectification framework that freezes the quantized backbone and trains a small classifier-head LoRA adapter using only source data. Recti-Q is architecture-agnostic across CNNs and Transformers, supports efficient teacher-free training, and recovers a significant portion of the lost robustness, in some cases matching or exceeding FP32 performance. At less than 1% parameter overhead (as small as 6 KB), Recti-Q preserves over 99% of PTQ memory savings, adds negligible compute, and enables low-bandwidth Over-The-Air (OTA) resilience patching for deployed robotic fleets operating in unpredictable physical environments.

I. INTRODUCTION

Visual classification is a core component of many robotic perception pipelines, including autonomous navigation, traffic-sign recognition, and object-centric decision making [1], [2], [3], [4]. These modules are commonly deployed on embedded and edge platforms with tight constraints on memory, latency, and energy [5], [6], making efficiency a deployment requirement. Meanwhile, robots operate in open environments where inputs are degraded by sensor noise, motion blur, precipitation, illumination changes, and other disturbances. For example, in an autonomous driving stack, a quantized perception module that correctly recognizes a stop sign in clear weather may fail under rain, lens splatter, or glare, and the error can propagate to downstream tracking, prediction, or planning. A perception model must therefore be not only accurate, but also robust to distribution shift. As shown in Figure 1, a full-precision model may handle such perturbations, while its quantized counterpart can fail on the same input.

Post-Training Quantization (PTQ) reduces model size while largely preserving clean accuracy and avoids costly end-to-end retraining, making it attractive for edge deployment [7], [6], [8]. However, quantization research typically emphasizes clean accuracy, compression, or throughput, while robustness-preserving alternatives often require full-data QAT or distillation, weakening PTQ’s deployment

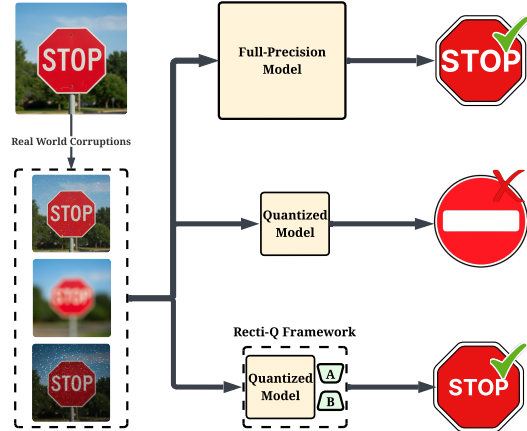


Fig. 1: **Quantization can severely degrade robustness under realistic perturbations.** A clean “STOP” sign is corrupted and passed to three visual classification systems representative of an edge robotic perception setting. The full-precision model predicts STOP (correct). The quantized model misreads it as No Entrance (wrong). Our framework, **Recti-Q**, keeps the quantized backbone and adds a lightweight adapter, restoring the correct STOP prediction while preserving efficiency.

advantage [7]. This motivates a critical question: *do PTQ models remain reliable under distribution shift?*

In this paper, we identify and quantify this gap, which we term the *Quantization-Induced Robustness Gap*: the loss in out-of-distribution (OOD) robustness caused by PTQ even when clean accuracy remains high. PTQ models can appear reliable on clean inputs (Table I) but, as shown in Figure 2, degrade sharply once the input distribution shifts. We evaluate on two complementary robustness benchmarks: ImageNet-C [9], which includes realistic corruptions (e.g., noise, blur, contrast change, weather-like effects), and PACS [10], which probes domain shift across visual styles. Using these foundational benchmarks, we isolate perception-level failure modes and show that Recti-Q addresses the underlying quantization-induced feature distortion.

To repair this gap, we propose **Recti-Q**, a lightweight framework that decouples compression from robustness. Recti-Q freezes the quantized backbone and adds a tiny, parameter-efficient adapter that rectifies quantization-induced feature perturbations at the classifier head. The adapter is trained only on source data, adds negligible parameter overhead, and supports low-bandwidth Over-The-Air (OTA) robustness patching for deployed robotic fleets by transmit-

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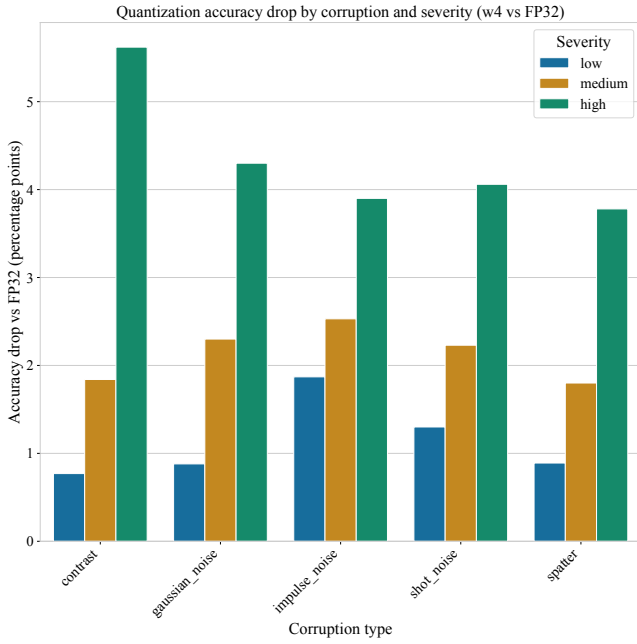


Fig. 2: **Quantization degrades robustness under distribution shift.** We plot the accuracy drop of a 4-bit quantized model (relative to its FP32 version) on ImageNet-C at severities 1, 3, and 5 (highest level). The robustness gap grows with corruption severity, showing that PTQ can significantly reduce reliability under realistic input degradation. This issue motivates our **Recti-Q** framework, which recovers a substantial portion of this lost robustness.

ting only the adapter parameters. Recti-Q is architecture-agnostic, working with both CNNs and Transformers, and can be trained with or without a full-precision teacher.

Our results show that Recti-Q recovers a substantial portion of the robustness lost to quantization on both ImageNet-C and PACS, while preserving the memory and efficiency benefits of 4-bit quantization. Our key contributions are:

- We identify and quantify the *Quantization-Induced Robustness Gap*, where PTQ models retain strong in-distribution accuracy yet degrade disproportionately under OOD corruption and domain shift.
- We introduce **Recti-Q**, a lightweight and architecture-agnostic framework that repairs a frozen quantized model using a parameter-efficient adapter, enabling low-bandwidth OTA robustness patching.
- We show via extensive experiments on ImageNet-C and PACS that Recti-Q substantially improves OOD robustness while retaining the efficiency benefits of 4-bit quantization.

II. RELATED WORK

A. Post-Training Quantization

Quantization enables deployment in resource-constrained settings by reducing the bit-width of weights and/or activations [7], which is especially important for on-board robotic perception under tight latency, power, and memory budgets. Post-Training Quantization (PTQ) is appealing because it is

applied after training, requires little or no fine-tuning data, and avoids full retraining [7]. As modern backbones and low-bit regimes become more challenging (e.g., ViTs [11] and 4-bit quantization), PTQ research has emphasized better calibration [12], [13], careful parameter choices (e.g., clipping/scales), and data-free/few-shot distillation to preserve clean accuracy [7]. In this work, we use a simple, uncalibrated PTQ scheme (e.g., `torchao` [14]) as a deployment baseline without a dedicated calibration dataset or weight fine-tuning; as shown by Table I, such PTQ pipelines can appear reliable on clean inputs, motivating closer evaluation under shift.

B. Robustness in Compressed Models

Efficiency-driven compression has motivated studies of robustness under adversarial perturbations and distribution shift. In embodied systems, quantization-induced feature instability (and, in detection stacks, unstable boxes/tracks) can propagate into downstream subsystems like modular trajectory prediction and motion planning, making robustness beyond clean accuracy a safety requirement. Prior work shows quantization can exacerbate adversarial vulnerability [15], [16], and robustness under practical shifts is often evaluated via common corruptions (ImageNet-C [9]) and domain generalization (e.g., PACS [10]). Robustness-preserving quantization methods frequently rely on Robust Quantization-Aware Training (QAT) or robust distillation [17], [16], which typically require full training data and expensive end-to-end retraining, weakening PTQ’s deployment appeal in robotics. Our results in Figure 2 reflect this gap even when clean accuracy is preserved (Table I), motivating a post-training robustness repair on a frozen quantized backbone.

C. Parameter-Efficient Fine-Tuning (PEFT) and Adapters

Recti-Q builds on Parameter-Efficient Fine-Tuning (PEFT) [18], where small auxiliary parameters adapt a frozen backbone. Adapter-based tuning includes bottleneck adapters and low-rank updates [19]; LoRA [20] is widely used, and PEFT has been applied to domain adaptation and transfer learning [21]. Related lightweight strategies include last-layer retraining for group robustness [22], [23].

Recent adapter work also targets constraint-induced errors in settings not directly comparable to ours. For example, Quadapter mitigates *GPT-2* quantization issues via activation-outlier handling [24], and other work studies adapting *quantized LLMs* [25]; these are perplexity-centric and offline, unlike real-time robotic perception under physical-world shift. RepairLLaMA targets *program repair* [26], and Robustness Feature Adapter improves efficiency in *adversarial training* [27], which differs from post-training robustness repair for PTQ vision models under ImageNet-C/PACS shifts.

In contrast, Recti-Q freezes the PTQ vision backbone and localizes adaptation to a small classifier-head module to preserve the efficiency profile and avoid architecture-specific integration. This minimal parameter footprint is well-suited for Over-The-Air (OTA) robustness patching in

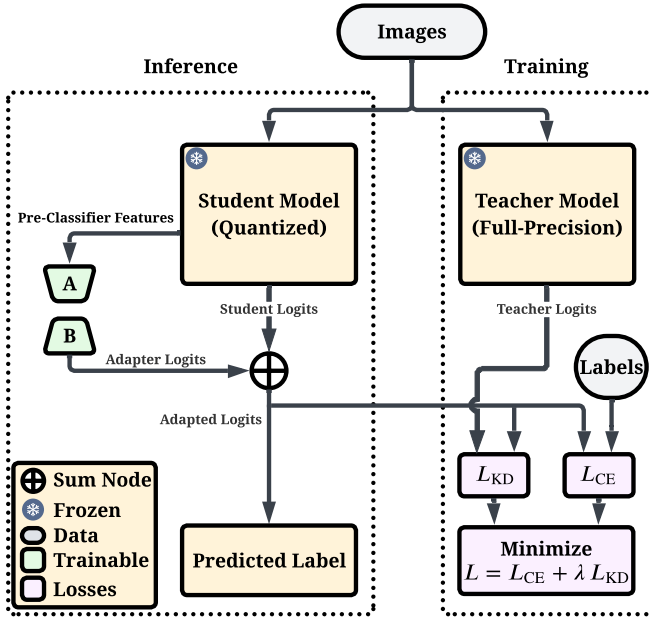


Fig. 3: **Recti-Q training and inference.** A frozen quantized student produces logits that are refined by a trainable LoRA adapter placed at the classifier head. The adapter applies a down-projection A and an up-projection B (scaled by α/r) to the pre-classifier features. Training uses only source-domain images and optimizes cross-entropy on labels and a distillation term that compares adapted student logits with teacher logits. The teacher is used only during training. At inference, the quantized backbone and the adapter run, preserving efficiency while restoring *robustness*.

TABLE I: **PTQ preserves ID accuracy.** We evaluate top-1 accuracy of FP32 models and their W4 quantized counterparts on standard ID test splits. On both PACS and ImageNet, the W4 models show negligible degradation. This establishes that standard ID metrics *alone* are insufficient to evaluate **model robustness**, as they fail to capture the substantial OOD degradation, to be shown in Section IV.

Model	PACS (ID) (%)		ImageNet (ID) (%)	
	FP32 Acc.	W4 Acc.	FP32 Acc.	W4 Acc.
ResNet50	94.99	94.87	80.38	80.31
DeiT-t	94.48	94.26	72.16	71.54
DeiT-s	96.38	96.38	79.85	78.94
DeiT-b	97.21	97.55	81.98	81.50

deployed robotic fleets, a deployment paradigm that remains underexplored in current robot learning literature.

III. METHOD

We start from a pretrained vision model f , which we quantize post-training (PTQ) to obtain an efficient, frozen backbone f_q . As shown in Table I, PTQ largely preserves standard ID accuracy. However, as we will show in Section IV, the same quantized models suffer a substantial drop in robustness on OOD data.

Our goal is to repair this OOD robustness without re-training the full model or incurring high computational costs. We propose Recti-Q, a framework that decouples compression from robustness recovery. The key idea is to freeze the efficient backbone f_q and train a single, parameter-efficient *robustness-repairing adapter*, g_ϕ , using only source-domain data.

A. Adapter Design: Rectifying Feature Perturbations

We hypothesize that PTQ does not destroy the model’s knowledge, but rather perturbs its internal feature representations and decision boundaries. In embodied settings, such unstable representations can translate into inconsistent predictions (and in detection-centric stacks, unstable boxes/tracks) that propagate into downstream autonomy modules. Therefore, simply re-weighting the final, damaged logits is insufficient. Our primary approach is to intervene at the feature level, just before the classifier head.

Classifier-LoRA Adapter (Recti-Q): We introduce a low-rank adapter, g_ϕ , that operates on the pre-classifier features, $u = \text{PRECLASSIFIERFEATURES}(f_q, x) \in \mathbb{R}^d$. This adapter is modeled as a standard LoRA module [20], consisting of a down-projection $A \in \mathbb{R}^{r \times d}$ and an up-projection $B \in \mathbb{R}^{C \times r}$ (with rank r and C classes):

$$g_\phi(u) = B(A(u)) \cdot (\alpha/r), \quad (1)$$

so $g_\phi(u) \in \mathbb{R}^C$ is a logit-space correction computed from pre-classifier features. The final logits are

$$z = z_q + g_\phi(u) = f_q(x) + g_\phi(u). \quad (2)$$

Equivalently, if the quantized head is $W_q \in \mathbb{R}^{C \times d}$ with $z_q = W_q u$, Recti-Q induces a low-rank update to the head, $z = (W_q + (\alpha/r)BA)u$, while keeping W_q frozen. Conceptually, this low-rank update acts as a rapid feature-space calibration, realigning representations that become distorted when the quantized weights process out-of-domain physical sensor inputs (e.g., camera glare or weather artifacts). Crucially, the up-projection B is zero-initialized, so at the start of training, $z = z_q$. This ensures a stable training process, where the adapter learns to gently correct the quantized model’s predictions.

B. Recti-Q Training and Inference

The complete Recti-Q framework is depicted in Figure 3. During training, the quantized student f_q is frozen. The full-precision teacher f_t is also kept frozen and is only required if knowledge distillation is applied ($\lambda > 0$). Our framework fully supports a “teacher-free” mode ($\lambda = 0$) for maximum training efficiency. In all cases, only the lightweight adapter g_ϕ is trained. The full training procedure is detailed in Algorithm 1.

At inference, the teacher model is discarded. The forward pass consists of a single pass through the frozen quantized backbone f_q and the tiny, trained adapter g_ϕ . This process, detailed in Algorithm 2, adds negligible parameter overhead while restoring OOD robustness.

TABLE II: **Recti-Q Recovers OOD Robustness (PACS) While Preserving Efficiency.** Our method (Ours) not only recovers the accuracy of the degraded W4 baseline but in some cases **exceeds the original FP32 performance** (e.g., DeiT-t on cartoon, 73.34% vs 73.08%). Crucially, the final three columns show the complete efficiency story: our total model size (‘Ours’) **retains over 99% of the W4 model’s memory savings**, demonstrating a clear, practical win for both **robustness** and **efficiency**.

Model	Domain	OOD Accuracy (%)			Robustness Delta (pp)		Model Size (MB)		
		FP32	W4	Ours	W4 vs FP32 (Gap)	Ours vs W4 (Recov.)	FP32	W4	Ours
DeiT-s	sketch	68.59	64.57	65.74	-4.02	+1.17	82.71	25.64	25.74
DeiT-t	cartoon	73.08	71.72	73.34	-1.36	+1.62	21.13	15.59	15.64
DeiT-t	art_painting	74.90	73.63	75.15	-1.27	+1.52	21.13	15.59	15.64
DeiT-s	art_painting	88.38	87.21	88.28	-1.17	+1.07	82.71	25.64	25.74
ResNet50	sketch	72.46	72.42	73.30	-0.04	+0.88	90.03	90.03	90.53

TABLE III: **Recti-Q Recovers OOD Robustness on ImageNet-C While Preserving Efficiency.** Our method (Ours) is applied to the W4 baseline on the most challenging corruptions from ImageNet-C (severity 5). The table demonstrates a strong and consistent recovery, **closing nearly 50% of the robustness gap** on the worst-case contrast corruption (recovering +2.68pp of a -5.62pp drop). Crucially, the final three columns show this **robustness** gain is achieved with minimal cost: the total model size (‘Ours’) **retains over 99% of the W4 model’s memory savings**, validating our framework’s practical efficiency.

Model	Corruptions	OOD Accuracy (%)			Robustness Delta (pp)		Model Size (MB)		
		FP32	W4	Ours	W4 vs FP32 (Gap)	Ours vs W4 (Recov.)	FP32	W4	Ours
DeiT-s	contrast	39.46	33.84	36.52	-5.62	+2.68	84.17	26.16	26.50
DeiT-s	gaussian_noise	33.18	28.88	29.30	-4.30	+0.42	84.17	26.16	26.50
DeiT-s	shot_noise	30.36	26.30	26.67	-4.06	+0.37	84.17	26.16	26.50
DeiT-s	impulse_noise	32.82	28.92	29.19	-3.90	+0.27	84.17	26.16	26.50
ResNet50	impulse_noise	19.83	19.68	27.04	-0.15	+7.37	97.79	91.02	91.77

C. Optimization Objective

Only adapter parameters are updated. The quantized backbone and, when used, the float teacher remain frozen and in evaluation mode. Training uses standard classification loss on allowed source data. When a float teacher is available, we add a temperature-scaled distillation term to preserve class structure learned by the original model [28].

Plainly, the objective is defined as

$$L = L_{CE} + \lambda L_{KD}, \quad (3)$$

where L_{CE} is the cross-entropy loss on the source labels, L_{KD} is the Kullback–Leibler divergence between the adapted and teacher predictions (with temperature T), and λ controls the strength of the distillation. In teacher-free runs, we set $\lambda = 0$. This provides a practical and highly efficient adaptation scenario that avoids the computational cost and memory overhead of loading and running the full-precision teacher model. We analyze the performance of this teacher-free approach in our experiments (see Section IV), demonstrating its efficacy as a minimal-cost solution.

We use the AdamW optimizer [29] with a small learning rate, light weight decay, and a short cosine schedule (converging in just a few epochs, typically requiring only minutes of numerical optimization on a standard consumer GPU or local edge server, facilitating rapid depot-level adaptation), with the same preprocessing and augmentations as in the source training recipe.

D. Source-only Adaptation Protocol

We follow leakage-free, community-standard evaluation. For ImageNet-C, adapters are trained only on a fixed 5% class-balanced subsample of ImageNet-1k train and validated on ImageNet-1k val; evaluation is on ImageNet-C across corruption types and severities [9]. For PACS, we use leave-one-domain-out: for each target domain, adapters are trained on the other three source domains, validated on source splits, and evaluated once on the held-out target [10]. No target images are used for training or model selection. This source-only constraint is practical and challenging, as it forgoes any target-domain data, a paradigm explored in other robustness frameworks [30]. In embodied AI, this protocol reflects “Lab-to-Wild” or “Sim-to-Real” deployment, where a robot must operate in unseen environments without prior target-domain calibration.

E. Complexity and Deployment

Recti-Q concentrates adaptation at the head, so the added parameter overhead is small relative to the backbone. For a classifier with feature dimension d and C classes, the LoRA path adds roughly $r(d + C)$ weights for rank r , much smaller than the full head. The backbone remains quantized and unchanged, so the total memory footprint is preserved while robustness is improved. This also enables low-bandwidth Over-The-Air (OTA) robustness patching for deployed robotic fleets: once a base quantized model is on-device, only the small adapter parameters need to be

Algorithm 1 Recti-Q training on a frozen quantized backbone with a classifier-head LoRA adapter and optional KD. In teacher-free runs, we set $\lambda = 0$ and skip computing s_t and L_{KD} in implementation.

Require: Frozen quantized model f_q ; optional frozen teacher f_t ; trainable LoRA adapter g_ϕ ; source data $\mathcal{D}_{src}$; validation data $\mathcal{D}_{val}$; distillation weight λ ; temperature T

```

1: for epoch = 1, ...,  $E$  do
2:   for minibatch  $(x, y)$  in  $\mathcal{D}_{src}$  do
3:      $z_q \leftarrow f_q(x)$ 
4:      $u \leftarrow \text{PRECLASSIFIERFEATURES}(f_q, x)$   $\triangleright u$  is
       cached from the same forward pass
5:      $z \leftarrow z_q + g_\phi(u)$ 
6:      $L_{CE} \leftarrow \text{CE}(z, y)$ 
7:      $s_z \leftarrow \text{softmax}(z/T)$ 
8:      $s_t \leftarrow \text{softmax}(f_t(x)/T)$ 
9:      $L_{KD} \leftarrow \text{KL}(s_t \| s_z) T^2$ 
10:     $L \leftarrow L_{CE} + \lambda L_{KD}$ 
11:    Update  $\phi$  with AdamW on  $L$ 
12:  end for
13:  Validate on  $\mathcal{D}_{val}$ ; keep the best  $\phi$ 
14: end for
15: return trained adapter  $g_\phi$ 

```

Algorithm 2 Inference with **Recti-Q**

Require: Frozen f_q , trained g_ϕ , input x

```

1:  $z_q \leftarrow f_q(x)$ 
2:  $u \leftarrow \text{PRECLASSIFIERFEATURES}(f_q, x)$   $\triangleright u$  is cached
   from the same forward pass
3:  $z \leftarrow z_q + g_\phi(u)$ 
4: return  $\hat{y} = \arg \max z$ 

```

transmitted to update the system for new environments. Because the adapter operates solely on the low-dimensional pre-classifier features, it requires only a trivial number of multiply-accumulate (MAC) operations ($r \times d$ and $C \times r$). Compared to the billions of MACs executed by the dense visual backbone, this sequential FLOP addition is asymptotically negligible, ensuring that the real-time inference latency (Hz) required for high-speed robotic control loops remains completely unaffected.

IV. RESULTS

A. Experimental Setup

Datasets and Protocols. Our experiments are conducted on two standard out-of-distribution (OOD) benchmarks, following strict source-only evaluation protocols.

- **ImageNet-C:** For corruption robustness, we use ImageNet-C [9]. We follow a resource-efficient, “leakage-free” protocol: adapters are trained on a fixed 5% class-balanced subsample of the ImageNet-1k *train* set and validated on the standard ImageNet-1k *val* split. All final results are reported on the ImageNet-C test set. No corrupted data is seen during training or validation.

- **PACS:** For domain generalization, we use the PACS dataset [10], which contains four distinct domains (Photo, Art, Cartoon, Sketch). We follow the standard leave-one-domain-out (LODO) protocol. For each domain held out as the target, we train adapters using only the data from the remaining three source domains.

Models and Baselines. We evaluate on both CNNs and Transformers to show broad applicability: ResNet-50, DeiT-Tiny (DeiT-t), DeiT-Small (DeiT-s), and DeiT-Base (DeiT-b). All models use pretrained weights from timm [31]. We compare three primary configurations:

- **FP32 (Reference):** The original, full-precision model, which serves as our “gold standard” for robustness.
- **PTQ (W4) (Problem Baseline):** The frozen backbone after applying weight-only 4-bit quantization using torchao. We use the Int4WeightOnlyConfig with use_hqq=True. This enables Half-Quadratic Quantization (HQQ), a simple, optimization-based technique that does not require a calibration dataset. This is a crucial design choice: by using a strong, calibration-free method, we establish a rigorous baseline and avoid introducing data-driven artifacts that could confound the OOD robustness evaluation. This model therefore represents a “pure” PTQ baseline and highlights the true “Quantization-Induced Robustness Gap.” We focus our main analysis on W4 as a strict stress test, since it exhibits the most severe robustness degradation; repairing this regime validates Recti-Q under aggressive compression. Furthermore, W4 simulates the extreme compression regimes increasingly demanded by memory-bandwidth-bound robotic edge accelerators (e.g., edge TPUs or embedded GPUs) deployed in SWaP-constrained autonomous systems.
- **Recti-Q (Ours):** The identical W4-quantized backbone augmented with our lightweight adapter. The backbone remains frozen, and only the adapter parameters are trained.

Implementation Details. Unless stated otherwise, all Recti-Q results use our Classifier-LoRA Adapter with a rank $r = 64$ and $\alpha = 16$ (applied with α/r scaling). We train all adapters using the AdamW optimizer [29] with a learning rate of 3×10^{-4} and weight decay of 1×10^{-4} . Adapters for ImageNet-C are trained for 5 epochs; adapters for PACS are trained for 5-10 epochs. We use a training batch size of 128 and a validation/test batch size of 256. For Knowledge Distillation (KD), we use the FP32 model as the teacher, a temperature $T = 4$, and a distillation weight $\lambda \in \{0, 0.5, 1.0\}$. The $\lambda = 0$ case represents our “teacher-free” variant, which relies solely on the cross-entropy loss.

B. The Quantization-Induced Robustness Gap

We begin our experimental analysis by investigating the effect of 4-bit Post-Training Quantization (W4 PTQ) on model robustness.

The ID Accuracy is Preserved, but Misleading. First, we establish a critical, but deceptive, baseline. We evaluate

TABLE IV: **PTQ substantially degrades OOD robustness on the PACS domain-shift benchmark.** We show the accuracy of the FP32 “gold standard” vs. the W4 PTQ baseline on held-out OOD domains. The table highlights the worst-case “Model-Domain” pairs, showing a significant drop in robustness (e.g., -4.02 pp for DeiT-s on sketch) that was not visible in the ID data (Table I).

OOD Accuracy Analysis				
Model	Domain	FP32 (%)	W4 (%)	W4 Drop (pp)
DeiT-s	sketch	68.59	64.57	-4.02
DeiT-t	cartoon	73.08	71.72	-1.36
DeiT-t	art_painting	74.90	73.63	-1.27
DeiT-s	art_painting	88.38	87.21	-1.17

our models on standard ID test sets. As shown in Table I, applying W4 PTQ results in a negligible accuracy change. On the ImageNet validation set, the accuracy drop is less than 1.0 percentage points for all DeiT models. On the PACS ID split, performance is nearly identical, with DeiT-b even showing a minor improvement. A developer looking only at these standard metrics would conclude the quantization was successful and robust to deploy.

The “Robustness Gap”: OOD Robustness Degrades. However, this perceived robustness shatters when the models are evaluated under out-of-distribution (OOD) shifts. This disparity is evident in two distinct OOD settings. For brevity, we report representative worst-case domains and corruptions here to highlight the phenomenon.

- **On Common Corruptions (ImageNet-C):** We first analyze performance on the ImageNet-C benchmark. Table V quantifies the accuracy drop for the worst-case corruptions at severity 5. For the DeiT-s model, this drop is severe, reaching -5.62 percentage points on the ‘contrast’ corruption. In an embodied context, this statistically significant drop translates directly to an increased rate of catastrophic perception failures, such as failing to detect an obstacle in low-light or foggy conditions, despite the model operating perfectly in a controlled lab environment.
- **On Domain Shift (PACS):** This robustness loss is not limited to corruptions. We observe a similar pronounced drop in the PACS domain-shift benchmark. Table IV quantifies this degradation on the four domains that exhibited the largest accuracy drop. On the DeiT-s sketch domain, W4 quantization causes the model to lose over 4 percentage points of accuracy, erasing a significant portion of its generalization capability.

Taken together, Table V and Table IV provide clear evidence of a Quantization-Induced Robustness Gap. While PTQ successfully preserves ID accuracy, it creates a severe and previously under-explored vulnerability to OOD shifts. This gap motivates the central goal of our work: to repair this robustness degradation.

TABLE V: **PTQ substantially degrades OOD robustness on ImageNet-C.** This table highlights the worst-case drops for a selection of corruptions at severity 5 (for DeiT-s). This severe drop was not visible in the ID data (Table I).

OOD Accuracy Analysis				
Model	Corruptions	FP32 (%)	W4 (%)	W4 Drop (pp)
DeiT-s	contrast	39.46	33.84	-5.62
DeiT-s	gaussian_noise	33.18	28.88	-4.30
DeiT-s	shot_noise	30.36	26.30	-4.06
DeiT-s	impulse_noise	32.82	28.92	-3.90

C. Recti-Q Recovers OOD Robustness

Having established the “Quantization-Induced Robustness Gap” in Section IV-B, we now demonstrate our method’s ability to repair it. We present our main results for both domain shift (PACS) in Table II and common corruptions (ImageNet-C) in Table III.

Performance on Domain Shift (PACS). The results on the PACS benchmark are compelling. As shown in Table II, our Recti-Q method not only recovers the accuracy drop but, in several cases, exceeds the original FP32 “gold standard” performance. For instance, on the cartoon and art_painting domains, the DeiT-t model, when augmented with our adapter, achieves 73.34% and 75.15% accuracy, surpassing the FP32 performance of 73.08% and 74.90%, respectively. These gains are best interpreted as calibration of quantization-induced shifts in feature geometry and the decision boundary: Recti-Q restores robustness supported by the frozen backbone rather than acquiring semantic understanding of unseen classes. Crucially, the final three columns quantify our core Recti-Q claim. The full FP32 DeiT-s model (82.71 MB) is compressed to just 25.64 MB by W4 PTQ. Our method adds only 0.10 MB (25.74 MB total), retaining over 99.6% of the memory savings while restoring, and even improving robustness.

Performance on Common Corruptions (ImageNet-C). We observe a similarly strong success on the challenging ImageNet-C benchmark, shown in Table III. Here, we focus on the most severe corruptions (severity=5) for the DeiT-s model.

The recovery is both significant and consistent. For the contrast corruption, which causes the largest drop of -5.62 pp, our method recovers +2.68 pp, closing nearly 50% of the robustness gap. While the recovery magnitude varies by corruption type, Recti-Q provides consistent, positive gains across all other tested corruptions, demonstrating a robust repair mechanism.

This robustness is again achieved with near-zero cost. The final three columns show the total model size (26.50 MB) remains almost identical to the W4 baseline (26.16 MB). Taken together, these results validate our central hypothesis: by decoupling compression and robustness, we can use a tiny, parameter-efficient adapter to repair the OOD robustness of a frozen quantized model, all while retaining over 99% of the memory savings as discussed in Section III.

TABLE VI: **Ablation: The FP32 teacher (KD) is not always required.** We compare the teacher-free ($\lambda = 0$) Recti-Q with two teacher-guided variants ($\lambda = 0.5$, $\lambda = 1.0$) on the worst-case PACS domains. The teacher-free model is highly competitive and even achieves the **best performance** for DeiT-t.

Model	Domain	Baseline	Ours (Accuracy %)		
		W4 Acc.	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1.0$
DeiT-s	sketch	64.57	64.65	65.49	65.74
DeiT-s	art_painting	87.21	88.28	87.68	87.71
DeiT-t	cartoon	71.72	73.34	72.87	72.85
DeiT-t	art_painting	73.63	75.15	74.41	74.24

D. Ablation Studies

We conduct ablation studies to validate our key design choices: the necessity of the FP32 teacher (KD loss), the superiority of our feature-space adapter, and the effect of adapter rank.

1) *Is the FP32 Teacher Necessary?*: A key component of our efficiency claim is the ability to train without a large, full-precision teacher model. We test this by comparing our full method (using KD with $\lambda > 0$) against a “teacher-free” variant that trains only with cross-entropy ($\lambda = 0$). We focus our analysis on the four worst-case domains from our main results. The results, shown in Table VI, show that for the DeiT-t model, the teacher-free ($\lambda = 0$) variant is the strongest performer, achieving the highest accuracy on both the art_painting (+1.52 pp gain) and cartoon (+1.62 pp gain) domains. For DeiT-s, the teacher provides a clear benefit on the sketch domain (+1.17 pp vs. +0.08 pp), but the teacher-free model is still highly effective on art_painting (recovering +1.07 pp).

This is an important finding: while the FP32 teacher can provide a boost in specific cases, our Recti-Q framework is highly effective even in its most efficient, teacher-free configuration. This confirms that our method is practical for scenarios where a teacher model is unavailable or too costly to run.

2) *Is Feature-Space Rectification Necessary?*: Our core hypothesis is that we must “rectify feature perturbations” (Section III-A). To prove this, we compare our Classifier-LoRA Adapter (which operates on features) against a Logit-Space Adapter (a simple MLP baseline that operates on the final, damaged logits).

As shown in Table VII, the baseline provides no measurable gain. On the DeiT-s sketch domain, the Logit-Space Adapter provides zero recovery (+0.00 pp) compared to the +1.17 pp gain from our feature-space method. This confirms our hypothesis: the OOD degradation is a deep feature-level problem, and simply re-weighting the final logits is an ineffective solution.

3) *Effect of Adapter Rank (r)*: Finally, we analyze the parameter-efficiency of Recti-Q by varying the LoRA rank r . We test this on the most challenging DeiT-s sketch domain, with results shown in Table VIII.

TABLE VII: **Ablation: Feature-Space (Ours) vs. Logit-Space (Baseline).** We compare our Recti-Q (Feature-LoRA) against a baseline MLP adapter on the final logits. The results, on the worst-case DeiT-s sketch domain, prove our feature-space rectification hypothesis is necessary.

Method	Accuracy (%)	Gain (pp)
W4 (Baseline)	64.57	-
+ Logit-Space Adapter	64.57	+0.00
+ Recti-Q (Ours)	65.74	+1.17

The data reveals a clear “sweet spot” of diminishing returns. A minimal rank of $r = 4$ (adding only 0.006 MB) recovers +0.77 pp, which is over 74% of the total gain achieved by the 16x larger $r = 64$ adapter. Increasing the rank to $r = 16$ (a mere 0.025 MB) captures nearly 90% of the maximum recovery (+0.95 pp).

This is a critical finding: it confirms that our method is not just parameter-efficient, but extremely parameter-efficient. At merely 6 KB (0.006 MB), this adapter payload can be transmitted near-instantaneously over low-bandwidth cellular or satellite networks, serving as concrete proof that Recti-Q provides a highly practical, near-zero-cost solution for Over-The-Air (OTA) robustness patching in remote robotic fleets.

TABLE VIII: **Ablation: Effect of LoRA rank (r) on robustness.** We analyze the accuracy vs. parameter cost on the worst-case DeiT-s sketch domain. The baseline W4 accuracy is 64.57%.

Rank (r)	Adapter Size (MB)	Accuracy (%)	Gain (pp)
4	0.006	65.34	+0.77
8	0.012	65.41	+0.84
16	0.025	65.52	+0.95
32	0.050	65.59	+1.02
64	0.100	65.61	+1.04

V. CONCLUSION

We addressed a critical, yet under-explored, trade-off between model efficiency and deployment robustness. We showed that standard Post-Training Quantization (PTQ), a vital tool for efficiency on edge platforms, can substantially degrade out-of-distribution (OOD) robustness, creating a “Quantization-Induced Robustness Gap.” We provided evidence for this gap on both common corruptions (ImageNet-C) and domain shifts (PACS), demonstrating that models that appear robust on in-distribution data can fail silently and severely under shift.

To address this, we introduced **Recti-Q**, a framework that decouples model compression from robustness recovery. Instead of expensive re-training, Recti-Q freezes the efficient, quantized backbone and trains a single, lightweight robustness-repairing adapter (< 1% overhead) using only source-domain data. Our approach is motivated by the hypothesis that PTQ perturbs the model’s feature space, which

can be corrected by a small, parameter-efficient adapter at the classifier head.

Our experiments validate this hypothesis. As shown in Table II and Table III, Recti-Q consistently recovers a significant portion of the robustness gap and, in some cases, even exceeds the original FP32 robustness, while retaining over 99% of the PTQ model’s memory savings and incurring negligible additional inference cost.

The implications of this work are practical. Recti-Q challenges the assumption that robust quantization necessarily requires computationally expensive Quantization-Aware Training. Instead, one can take an “off-the-shelf” quantized model and apply a rapid, low-cost robustness patch, enabling low-bandwidth Over-The-Air (OTA) updates for deployed fleets.

Limitations and scope. Recti-Q corrects quantization-induced feature distortion but cannot reconstruct information lost by the frozen backbone. Because the adapter acts only on the backbone’s output features, its effectiveness depends on those features retaining usable class structure; under sufficiently aggressive quantization, this assumption may fail and head-only rectification may become ineffective. Our evaluation is limited to closed-set covariate and domain/style shifts (ImageNet-C and PACS); Recti-Q has not been evaluated on semantic or open-set OOD and should not be expected to recognize categories outside the original training taxonomy.

REFERENCES

- [1] L. Zheng, H. Y. Araghi, T. Wu, S. Thalapanane, T. Zhou, and M. C. Lin, “Quantifying and modeling driving styles in trajectory forecasting,” 2025. [Online]. Available: <https://arxiv.org/abs/2503.04994>
- [2] K. Rajan and A. Saffiotti, “Towards a science of integrated ai and robotics,” *Artificial Intelligence*, vol. 247, pp. 1–9, 2017, special Issue on AI and Robotics. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0004370217300310>
- [3] P. Ganesan, “Revolutionizing robotics with ai,” *Machine Learning, and Deep Learning: A Deep Dive into Current Trends and Challenges. J Artif Intell Mach Learn & Data Sci*, vol. 1, no. 4, pp. 1124–1128, 2023.
- [4] L. W. Lee, A. Dabirian, I. P. McCarthy, and J. Kietzmann, “Making sense of text: artificial intelligence-enabled content analysis,” *European Journal of Marketing*, vol. 54, no. 3, pp. 615–644, 02 2020. [Online]. Available: <https://doi.org/10.1108/EJM-02-2019-0219>
- [5] C. Tanama, K. Peng, Z. Marinov, R. Stiefelwagen, and A. Roitberg, “Quantized distillation: Optimizing driver activity recognition models for resource-constrained environments,” in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2023, pp. 5479–5486.
- [6] O. Grainge, M. Milford, I. Bodala, S. D. Ramchurn, and S. Ehsan, “Design space exploration of low-bit quantized neural networks for visual place recognition,” *IEEE Robotics and Automation Letters*, vol. 9, no. 6, pp. 5070–5077, 2024.
- [7] F. Chen, Z. Luo, L. Zhou, X. Pan, and Y. Jiang, “Comprehensive survey of model compression and speed up for vision transformers,” *arXiv preprint arXiv:2404.10407*, 2024.
- [8] Z. Liu, Y. Wang, K. Han, W. Zhang, S. Ma, and W. Gao, “Post-training quantization for vision transformer,” *Advances in Neural Information Processing Systems*, vol. 34, pp. 28 092–28 103, 2021.
- [9] D. Hendrycks and T. Dietterich, “Benchmarking neural network robustness to common corruptions and perturbations,” 2019. [Online]. Available: <https://arxiv.org/abs/1903.12261>
- [10] D. Li, Y. Yang, Y.-Z. Song, and T. M. Hospedales, “Deeper, broader and artier domain generalization,” in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, Oct 2017.
- [11] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, “An image is worth 16x16 words: Transformers for image recognition at scale,” 2021. [Online]. Available: <https://arxiv.org/abs/2010.11929>
- [12] M. Nagel, M. van Baalen, T. Blankevoort, and M. Welling, “Data-free quantization through weight equalization and bias correction,” 2019. [Online]. Available: <https://arxiv.org/abs/1906.04721>
- [13] I. Hubara, M. Courbariaux, D. Soudry, R. El-Yaniv, and Y. Bengio, “Quantized neural networks: Training neural networks with low precision weights and activations,” 2016. [Online]. Available: <https://arxiv.org/abs/1609.07061>
- [14] A. Or, J. Liu, Y.-A. Li, Y.-H. Tang, H. Zuo, J. Yang, C. Shen, H. Huang, W.-C. Wu, P. Tang, P. D’Cunha, X. Jia, R. Raskar *et al.*, “TorchAO: Pytorch-native training-to-serving model optimization,” *arXiv preprint arXiv:2507.16099*, 2025, published in CODEML@ICML2025. [Online]. Available: <https://arxiv.org/abs/2507.16099>
- [15] Y. Liu, W. Zhang, and J. Wang, “Zero-shot adversarial quantization,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 1512–1521.
- [16] M. Goldblum, L. Fowl, S. Feizi, and T. Goldstein, “Adversarially robust distillation,” in *Proceedings of the AAAI conference on artificial intelligence*, vol. 34, no. 04, 2020, pp. 3996–4003.
- [17] B. Chmiel, R. Banner, G. Shomron, Y. Nahshan, A. Bronstein, U. Weiser *et al.*, “Robust quantization: One model to rule them all,” *Advances in neural information processing systems*, vol. 33, pp. 5308–5317, 2020.
- [18] Y. Xin, J. Yang, S. Luo, Y. Du, Q. Qin, K. Cen, Y. He, Z. Zhang, B. Fu, X. Yang, G. Zhai, M.-H. Yang, and X. Liu, “Parameter-efficient fine-tuning for pre-trained vision models: A survey and benchmark,” 2025. [Online]. Available: <https://arxiv.org/abs/2402.02242>
- [19] Adapter-based tuning. Emergent Mind. [Online]. Available: <https://www.emergentmind.com/topics/adapter-based-tuning>
- [20] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “Lora: Low-rank adaptation of large language models,” 2021. [Online]. Available: <https://arxiv.org/abs/2106.09685>
- [21] M. Rawhani, D. Karaboğa, U. Nalbantoglu, A. Baştürk, and B. Akay, “Efficient unsupervised domain adaptation with peft combinations,” in *2024 9th International Conference on Computer Science and Engineering (UBMK)*, 2024, pp. 169–174.
- [22] E. Z. Liu, B. Haghighi, A. S. Chen, A. Raghunathan, P. W. Koh, S. Sagawa, P. Liang, and C. Finn, “Just train twice: Improving group robustness without training group information,” 2021. [Online]. Available: <https://arxiv.org/abs/2107.09044>
- [23] M. Ghaznavi, H. Asadollahzadeh, H. Y. Araghi, F. H. Noohdani, M. H. Rohban, and M. S. Baghshah, “Annotation-free group robustness via loss-based resampling,” *arXiv preprint arXiv:2312.04893*, 2023.
- [24] M. Park, J. You, M. Nagel, and S. Chang, “Quadapter: Adapter for gpt-2 quantization,” in *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022, pp. 2510–2517.
- [25] A. Shen, Z. Lai, Q. Wang, X. Li, L. Zhang, D. Li, and J. Li, “Accurate and efficient fine-tuning of quantized large language models through optimal balance in adaptation,” *Transactions of the Association for Computational Linguistics*, vol. 13, pp. 861–877, 2025.
- [26] A. Silva, S. Fang, and M. Monperrus, “Repairllama: Efficient representations and fine-tuned adapters for program repair,” *IEEE Transactions on Software Engineering*, 2025.
- [27] Q. Wu, J. Guo, W. Wang, and Y. Wang, “Robustness feature adapter for efficient adversarial training,” *arXiv preprint arXiv:2508.17680*, 2025.
- [28] G. Hinton, O. Vinyals, and J. Dean, “Distilling the knowledge in a neural network,” 2015. [Online]. Available: <https://arxiv.org/abs/1503.02531>
- [29] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” 2019. [Online]. Available: <https://arxiv.org/abs/1711.05101>
- [30] A. H. Saberi, A. Najafi, A. Heidari, M. H. Movasaghinia, A. Motahari, and B. H. Khalaj, “Out-of-domain unlabeled data improves generalization,” *arXiv preprint arXiv:2310.00027*, 2023.
- [31] R. Wightman, “Pytorch image models (timm),” <https://github.com/rwightman/pytorch-image-models>, 2019.